

Dinghao Luo

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EDUCATION

Max Planck Florida Institute for Neuroscience / FAU, Jupiter, FL, USA

May 2021–Dec 2026

PhD candidate, mechanisms of episodic memory

- Advisor: Yingxue Wang, Ph.D.
- International Max Planck Research School (IMPRS) fellow
- Defence scheduled mid-October 2026

University of Cambridge, Cambridge, UK

Sept 2017–June 2020

First-Class Honours graduate, BA Biological Natural Sciences at Magdalene College

- Jardine Scholar, fully funded by Jardine Matheson on a merit-based selection
- First-Class graduate, awarded the Bundy Scholarship and the College Prize for Natural Sciences
- Magdalene Scholar, elected based on academic performance

The Affiliated High School of South China Normal University, Guangzhou, CN

Aug 2014–June 2017

Weighted GPA 4.81, ranked first among 131 students

RESEARCH

Graduate thesis, *PI: Dr. Yingxue Wang*,

Max Planck Florida Institute for Neuroscience, Jupiter, FL, USA

May 2021–Dec 2026

- Funded by the International Max Planck Research School program
- Research aim: to understand the moment-to-moment dopaminergic modulation of hippocampus-dependent goal-directed navigation
 - ▶ Performed extracellular recording with silicon probes, optogenetic experiments, dual-colour two-photon imaging with Ca^{2+} , dopamine, and norepinephrine sensors, and computational modelling to probe the recently confirmed locus coeruleus–hippocampal CA1 pathway, with a virtual reality goal-directed navigation task
 - ▶ Found that phasic locus coeruleus dopamine modulates CA1 population dynamics to sharpen goal-directed navigation, with phasic stimulation at navigation onset enhancing performance accuracy
 - ▶ Developed FibreSight, a U-Net-based segmentation tool with a Qt GUI for automated axon ROI identification in two-photon imaging data
 - ▶ Built end-to-end data pipelines in Python for large-scale electrophysiology and two-photon imaging analysis, including population analysis, sensor signal extraction, complete with behavioural event alignment

Undergraduate thesis and summer internship, *PI: Dr. Ole Paulsen*,

Department of Physiology, Development and Neuroscience, University of Cambridge, Cambridge, UK

July 2019–April 2020

- Research aim: to investigate whether dopamine is essential for a novel LTP induction protocol
 - ▶ Used whole-cell patch clamping to study how hippocampal CA1 long-term potentiation rules change with dopamine
 - ▶ Found that dopamine reduced the stimulation intensity required for long-term potentiation
 - ▶ Collected and analysed electrophysiological data, prepared and defended an undergraduate thesis ('Dopamine Facilitation of CA3–CA1 Behavioural Time-Scale Plasticity'), receiving a First-Class mark

Summer internship, *PI: Dr. Elizabeth Fisher*

Queen Square Institute of Neurology, University College London, London, UK

July 2018–August 2018

- Funded by the Genes and Development Summer Studentship of Genetics Society
- Research aim: to determine the causal genes for activity deficits of cathepsin B in amyloid pathology in people with Down Syndrome
 - ▶ Used Western Blot to quantify active cathepsin B, pro-cathepsin B and S100B
 - ▶ Performed protein activity assays for cathepsin B activity in mouse models of Down Syndrome

Research student, *PI: Dr. Anne Fontana*,

Syracuse University, Syracuse, NY, USA

February 2016–May 2016

- Organised by the Pioneer Academics Research Program
- Research aim: to design an experiment comparing lithium, valproate and aripiprazole for the treatment of bipolar disorder
 - ▶ Reviewed literature on the categorisation, time course, correlational factors, and treatments of bipolar disorder, focussing on lithium, valproate and aripiprazole
 - ▶ 'Comparison of the Effectiveness of Lithium, Valproate and Aripiprazole for Bipolar Disorder' was nominated for publication in the 2016 edition of *Pioneer Research Journal*

SELECTED PROJECTS

FibreSight [github.com/dinghaoluo/fibre-sight]

- A fully-packaged, click-to-run tool for segmenting neuronal fibres in two-photon calcium imaging data
- Trained a high-performance U-Net network with recall-biased operating point (F2 over Dice), Qt-based workbench GUI covering labelling, training, and prediction workflows

Kairos: *What are the conditions of scientific discovery?* [github.com/dinghaoluo/kairos]

- Scientometric analysis investigating whether field-state features predict paradigm shifts in science, inspired by the Park *et al.* (*Nature*, 2023) debate on whether scientific breakthroughs are less frequent in our times
- Built a data pipeline using the OpenAlex API, engineered temporal-leakage-safe features, and trained classification and survival models

SELECTED PUBLICATIONS

Luo, D., Jingyu C., Lin T., & Yingxue W. Locus coeruleus dopaminergic control of prospective dynamics in the hippocampus. Submitted.

Wu, Y., Mumford, P., Noy, S., Cleverley, K., Mrzyglod, A., **Luo, D.** *et al.* Cathepsin B abundance, activity and microglial localisation in Alzheimer's disease—Down syndrome and early onset Alzheimer's disease: the role of elevated cystatin B. *Acta Neuropathol. Commun.* 11, 132 (2023). <https://doi.org/10.1186/s40478-023-01632-8>.

SELECTED PRESENTATIONS

Locus coeruleus dopamine rapidly modulates CA1 dynamics to sharpen goal-directed navigation. COSYNE (Computational and Systems Neuroscience), Lisbon, Portugal, 12–17 March 2026.

Locus coeruleus dopaminergic modulation of CA1 shapes behavioural timescale dynamics for time integration. Lake Conference, Seattle, WA, USA, 28 September–2 October 2025.

Locus coeruleus regulates hippocampus-dependent integration at single-trial level. Memory Formation Forum, Max Planck Florida Institute for Neuroscience, Jupiter, FL, USA, 18 September 2024.

An LC–CA1 circuit for initiating path integration. Sunposium, West Palm Beach, FL, USA, 6 March 2024.

The effect of three copies of 40 gene orthologs of chromosome 21 on amyloid pathology in a mouse model. Alzheimer's Research UK Research Conference, Harrogate, UK, 19–20 March 2019. Presented by Paige Mumford.

Mapping the causal gene for cathepsin B activity deficits in DS–AD. Genetics Society Summer Workshop, Edinburgh, UK, 27–30 August 2018.

SCIENCE COMMUNICATION

Translator

Sheldrake, M. *Entangled Life* (Random House, 2020). Chinese translation (《菌络万象》). 后浪出版, 2024. Won Pingshan Book Award 2025.

Writer, editor, proofreader and translator, *Neu-reality* (神经现实),

Chinese education platform for neuroscience and psychology [<https://neu-reality.com>] July 2017–December 2021

- Founded and led the 'Neuro-Frontier' (神经前研) section, with sections such as the fortnightly 'Paper Alerts', 'Perspectives'/'Deep-dives'
- Wrote original articles, translated content with authorisation from *Quanta*, *Nautilus*, and *Aeon* magazines, edited and proofread original articles and translations for publication on the platform

CERTIFICATIONS

SQL Associate, *DataCamp* [SQA0011475257484], certified 22 July 2026

OTHER INFORMATION

Languages

- Fluent in Cantonese (native), Mandarin (native), English
- Conversational in German and Spanish

Programming languages

- Python (primary; scikit-learn, PyTorch, pandas, NumPy, SciPy, statsmodels, matplotlib, spaCy)
- SQL (PostgreSQL)
- R (tidyverse, survival, MatchIt)
- MATLAB

Musical work

- Composed, recorded and self-published 1 full-length album and 1 extended play on Bandcamp [<https://amoxitoxin.bandcamp.com>]